2011 F=ma Exam: Problem 20

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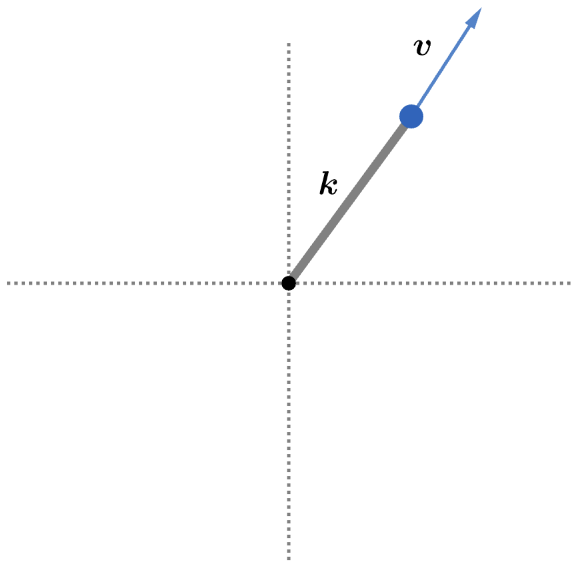

From the previous problem, we have the spring constant $k = 8\,\mathrm{N/m}$. The particle is launched with speed $v = \sqrt{(3\,\mathrm{m/s})^2 + (4\,\mathrm{m/s})^2} = 5\,\mathrm{m/s}$. By conservation of energy,

$$\frac{1}{2}mv^2 = \frac{1}{2}kA^2$$

$$A = v\sqrt{\frac{m}{k}} = (5\,\mathrm{m/s})\sqrt{\frac{2\,\mathrm{kg}}{8\,\mathrm{N/m}}} = 2.5\,\mathrm{m}$$

so the answer is $\boxed{\mathrm{B}}$.